EXPERIMENT 6

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Subject Name: Database Management System **Subject Code: 24CSH-298**

Experiment 6 – Understanding and Implementing Cursors for Row-by-Row Data Processing (Oracle, Accenture, Scaler, D. E. Shaw & Co)

Experiment

Experiment 6: Working with PL/SQL cursors to process multiple rows of a table individually. This experiment demonstrates the use of implicit cursors, explicit cursors, and cursor attributes to implement business logic on row-by-row data in Oracle and PostgreSQL.

Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Objective

- To implement and analyze implicit cursors for single-row operations.
- To implement explicit cursors to fetch and process multiple records.
- To apply cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN) for controlling program flow.
- To process employee records row by row and apply business logic effectively.

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

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Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

Problem Statement

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules. Using cursors allows row-by-row processing for detailed business logic implementation.

Practical / Experiment Steps

1. Design PL/SQL programs that demonstrate:
    - The use of implicit cursors for single-row DML operations
    - The use of explicit cursors to fetch and process multiple records
    - The application of cursor attributes to control program execution
2. Create PL/SQL programs to:
    - Fetch employee records from a database table using cursors
    - Process each record individually
    - Display results or apply business logic using cursor attributes

Procedure

1. Open Oracle SQL Developer or pgAdmin and connect to the database.
2. Create an employee table with sample data (id, name, department, salary).
3. Write an anonymous PL/SQL block demonstrating implicit cursor operations.
4. Write a PL/SQL block using an explicit cursor to fetch and process multiple records.
5. Apply cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN to control execution flow.
6. Execute the blocks and observe the output.

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7. Capture results and screenshots for documentation.

Input / Output Details

Input

employee table:

- id (Integer)
- name (Varchar)
- department (Varchar)
- salary (Integer)

Cursor logic used:

- Implicit cursor for single-row processing
- Explicit cursor for multi-row processing
- Cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN)

Step-wise Output

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Step 1 – Create employee Table

[SQL Worksheet]*

```
CREATE TABLE EMPLOYEE (
    EMP_ID NUMBER PRIMARY KEY,
    EMP_NAME VARCHAR2(50),
    SALARY NUMBER
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Table EMPLOYEE dropped.

Elapsed: 00:00:00.035

SQL> CREATE TABLE EMPLOYEE (
         EMP_ID NUMBER PRIMARY KEY,
         EMP_NAME VARCHAR2(50),
         SALARY NUMBER...
Show more...


Table EMPLOYEE created.

Elapsed: 00:00:00.012
```

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Step 2 – Insert Data into employee Table

[SQL Worksheet]*

```sql
CREATE TABLE EMPLOYEE (
    EMP_ID NUMBER PRIMARY KEY,
    EMP_NAME VARCHAR2(50),
    SALARY NUMBER
);
INSERT INTO EMPLOYEE VALUES (101, 'Amit', 25000);
INSERT INTO EMPLOYEE VALUES (102, 'Riya', 32000);
INSERT INTO EMPLOYEE VALUES (103, 'Rahul', 45000);
INSERT INTO EMPLOYEE VALUES (104, 'Sneha', 28000);
INSERT INTO EMPLOYEE VALUES (105, 'Arjun', 50000);

COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
1 row inserted.

Elapsed: 00:00:00.002

SQL> COMMIT


Commit complete.

Elapsed: 00:00:00.002
```

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Step 3 – Implicit Cursor Operation

[SQL Worksheet]*

```
BEGIN
    UPDATE EMPLOYEE
    SET SALARY = SALARY * 1.10
    WHERE SALARY < 30000;

    DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || SQL%ROWCOUNT);

    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Update Successful');
    END IF;
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> BEGIN
         UPDATE EMPLOYEE
         SET SALARY = SALARY * 1.10
         WHERE SALARY < 30000;...
Show more...


Rows Updated: 2
Update Successful


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.091
```

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Step 4 – Explicit Cursor Declaration and Fetch

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```
DECLARE
    CURSOR emp_cursor IS
        SELECT EMP_ID, EMP_NAME, SALARY FROM EMPLOYEE
        WHERE SALARY > 30000;

    v_id EMPLOYEE.EMP_ID%TYPE;
    v_name EMPLOYEE.EMP_NAME%TYPE;
    v_salary EMPLOYEE.SALARY%TYPE;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_salary;
        EXIT WHEN emp_cursor%NOTFOUND;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> DECLARE
        CURSOR emp_cursor IS
            SELECT EMP_ID, EMP_NAME, SALARY FROM EMPLOYEE
            WHERE SALARY > 30000;...
Show more...


ID: 102 Name: Riya Salary: 32000
ID: 103 Name: Rahul Salary: 45000
ID: 104 Name: Sneha Salary: 30800
ID: 105 Name: Arjun Salary: 50000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.079
```

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Step 5 – Process Records Using Cursor Attributes

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[SQL Worksheet]*

```
DECLARE
    CURSOR emp_cursor IS SELECT * FROM EMPLOYEE;
    emp_record EMPLOYEE%ROWTYPE;

BEGIN
    OPEN emp_cursor;

    IF emp_cursor%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor Opened Successfully');
    END IF;

    LOOP
        FETCH emp_cursor INTO emp_record;
        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'Employee: ' || emp_record.EMP_NAME ||
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
...
Show more...

Cursor Opened Successfully
Employee: Amit | Salary: 27500
Employee: Riya | Salary: 32000
Employee: Rahul | Salary: 45000
Employee: Sneha | Salary: 30800
Employee: Arjun | Salary: 50000
Total Rows Processed: 5


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.020
```

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Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the role of cursors in PL/SQL for handling multi-row query results.
- Differentiate between implicit cursors and explicit cursors.
- Use cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN.
- Develop PL/SQL programs that process database records row by row.

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- Apply cursor-based logic in real-world business scenarios used in organizations like Oracle, Accenture, Scaler, and D. E. Shaw & Co.